

RAHUL MITRA

OPERATIONS RESEARCH ENGINEER PROFILE

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PROFILE SUMMARY

Early-career operations research engineer combining NUS ISE foundations with deployable software, simulation, optimization, and analytics automation for scheduling, digital twin, process improvement, and decision-support roles.

EDUCATION

National University of Singapore	Aug 2023 - Jul 2027
- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics - Relevant coursework: Operations Research; Simulation; Statistics for Engineering Applications; Industrial Systems Engineering; Artificial Intelligence; 3D Computer Vision; Software Engineering - Certificate of Outstanding Performance in 3D Computer Vision (CS4277) - top student in a class of 24, NUS School of Computing.	
Anderson Serangoon Junior College	Feb 2019 - Dec 2020
- Singapore-Cambridge GCE A-Level	

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd	Jun 2026 - Present
- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.	
Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd	Jan 2026 - Jun 2026
- Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules. - Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.	
Base Support Assistant, Republic of Singapore Navy	Jan 2022 - Jan 2023
- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment. - Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.	
Software Engineer (Citizen Developer), People's Association	Sep 2025 - Present
- Built an end-to-end digital garden-plot balloting platform (admin and public-facing frontends, backend, GIS mapping, and deployment), now production-ready for national rollout across Singapore. - Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks. - Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders; awarded the S\$20,000 Sparks Community Innovation Fund.	

SELECTED PROJECTS AND COMPETITIONS

Hybrid Flow Shop Digital Twin Optimizer	2026
- Modeled production scheduling as a digital twin simulation and constraint-programming optimization problem, connecting ISE methods with deployment-oriented Python tooling.	
Volt Pulse SG - SMU Hack For Cities Finalist	Jan 2026
- Reached Top 8 of 50+ teams with an agentic AI platform for HDB energy-cost tracking using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.	
IE2110 Operations Research Graph Optimization	2024
- Applied IE2110 Operations Research I concepts from linear programming, network flow models, and nonlinear programming; implemented Jupyter SSSP/APSP graph optimization for engineering decision support.	

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Constraint Programming; Scheduling Optimization; Hybrid Flow Shop; Digital Twin Simulation; Dijkstra; SSSP/APSP; Python; Jupyter; Statistics; Simulation; Excel VBA; Decision Analytics.
- Excel 2019 Advanced - analytics automation and spreadsheet modeling. - NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows. - Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre	Sep 2025 - Present
- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.	
Astronomy Head and Robotics Member, STEM Inc. (ASRJC)	2019 - 2020
- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.	